Topic: Gradient

I. Scalar Functions

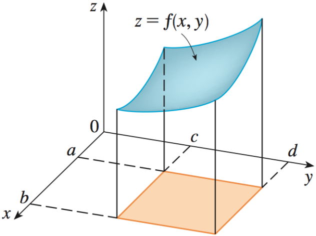
Defn: A Scalar Function is a function whose output is a scalar.

Ex: $f(x) = x^2$ is a scalar function since $f(2) = (2)^2 = 4 \longleftarrow$ (*scalar*)

$x \in \mathbf{R}$ and $f(x) \in \mathbf{R}$ Easy to Visualize

Ex: $f(x,y) = x - x\sqrt{y}$ is a scalar function since $f(1,4) = (1) - (1)\sqrt{(4)} = -1 \longleftarrow$ (*scalar*)

$(x,y) \in \mathbf{R}^2$ and $f(x,y) \in \mathbf{R}$ Easy to Visualize



Ex: $f(x,y,z) = \frac{\cos(z)}{x} - x$ is a scalar function since $f(-1,0,\pi) = \frac{\cos(\pi)}{(-1)} - (-1) = 2 \longleftarrow$ (*scalar*)

$(x,y,z) \in \mathbf{R}^3$ and $f(x,y,z) \in \mathbf{R}$ Difficult to Visualize (4D)

II. Gradient

<u>Defn</u>: Let D_f be the domain of the scalar function $f(x,y)$ and D_g be the domain of the scalar function $g(x,y,z)$. If f is differentiable on D_f and g is differentiable on D_g then the **gradient** of $f(x,y)$ and $g(x,y,z)$ is given by

$$\nabla f = \left[\frac{\partial f}{\partial x}, \frac{\partial f}{\partial y}\right] = \left[f_x, f_y\right] = f_x\mathbf{i} + f_y\mathbf{j}$$

$$\nabla g = \left[\frac{\partial g}{\partial x}, \frac{\partial g}{\partial y}, \frac{\partial g}{\partial z}\right] = \left[g_x, g_y, g_z\right] = g_x\mathbf{i} + g_y\mathbf{j} + g_z\mathbf{k}$$

<u>Ex</u>: $f(x,y) = xy + y$. Find ∇f

$$\frac{\partial f}{\partial x} = y \quad \text{and} \quad \frac{\partial f}{\partial y} = x + 1 \quad \Longrightarrow \quad \nabla f = [y, x+1]$$

<u>Ex</u>: $f(x,y,z) = z\ln(xy^2 + y)$. Find ∇f at the point $P(0,1,2)$

$$\frac{\partial f}{\partial x} = z\frac{1}{xy^2+y} \cdot \frac{\partial}{\partial x}(xy^2+y) = z\frac{1}{xy^2+y} \cdot y^2 = \frac{zy^2}{xy^2+y}$$

$$\frac{\partial f}{\partial y} = z\frac{1}{xy^2+y} \cdot \frac{\partial}{\partial y}(xy^2+y) = z\frac{1}{xy^2+y} \cdot (2xy+1) = \frac{z(2xy+1)}{xy^2+y}$$

$$\frac{\partial f}{\partial z} = \ln(xy^2+y)$$

$$\Longrightarrow \quad \nabla f(P) = \left[\frac{\partial f}{\partial x}(P), \frac{\partial f}{\partial y}(P), \frac{\partial f}{\partial z}(P)\right] = \left[\frac{(2)(1)^2}{(0)(1)^2+(1)}, \frac{(2)(2(0)(1)+1)}{(0)(1)^2+(1)}, \ln((0)(1)^2+(1))\right] = [2,2,0]$$

Prove the following two properties of gradients: Recall: $\nabla f = \left[\frac{\partial f}{\partial x}, \frac{\partial f}{\partial y}, \frac{\partial f}{\partial z}\right]$

1. $\nabla(f+g) = \nabla f + \nabla g$

$$
\begin{aligned}
\textit{Proof}: \quad \nabla(f+g) &= \left[\frac{\partial}{\partial x}(f+g), \frac{\partial}{\partial y}(f+g), \frac{\partial}{\partial z}(f+g)\right] && (\textit{Defn of Gradient}) \\
&= \left[\frac{\partial f}{\partial x} + \frac{\partial g}{\partial x}, \frac{\partial f}{\partial y} + \frac{\partial g}{\partial y}, \frac{\partial f}{\partial z} + \frac{\partial g}{\partial z}\right] && (\textit{Addition Rule} \times 3) \\
&= \left[\frac{\partial f}{\partial x}, \frac{\partial f}{\partial y}, \frac{\partial f}{\partial z}\right] + \left[\frac{\partial g}{\partial x}, \frac{\partial g}{\partial y}, \frac{\partial g}{\partial z}\right] && (\textit{Vector Addition}) \\
&= \nabla f + \nabla g && (\textit{Defn of Gradient} \times 2)
\end{aligned}
$$

2. $\nabla(f \cdot g) = \nabla f \cdot g + f \cdot \nabla g$

$$
\begin{aligned}
\textit{Proof}: \quad \nabla(fg) &= \left[(f\cdot g)_x, (f\cdot g)_y, (f\cdot g)_z\right] && (\textit{Defn of Gradient}) \\
&= \left[f_x \cdot g + f\cdot g_x, f_y\cdot g + f\cdot g_y, f_z\cdot g + f\cdot g_z\right] && (\textit{Product Rule} \times 3) \\
&= \left[f_x\cdot g, f_y\cdot g, f_z\cdot g\right] + \left[f\cdot g_x, f\cdot g_y, f\cdot g_z\right] && (\textit{Vector Addition}) \\
&= \left[f_x, f_y, f_z\right]\cdot g + f\cdot\left[g_x, g_y, g_z\right] && (\textit{Scalar} \cdot \textit{Vector}) \\
&= \nabla f \cdot g + f \cdot \nabla g && (\textit{Defn of Gradient} \times 2)
\end{aligned}
$$

III. Additional Concepts

Thm: $\nabla f(P)$ points in the direction of maximum increase of f at the point P.

Defn: The **Directional Derivative** of a function f, $D_{\mathbf{u}}f$, in the direction of the unit vector $\mathbf{u}$ is

$$D_{\mathbf{u}}f = \mathbf{u} \cdot \nabla f \qquad \text{so we also have} \qquad D_{\mathbf{u}}f = \mathbf{u} \cdot \nabla f = |\mathbf{u}||\nabla f| \cos\theta$$

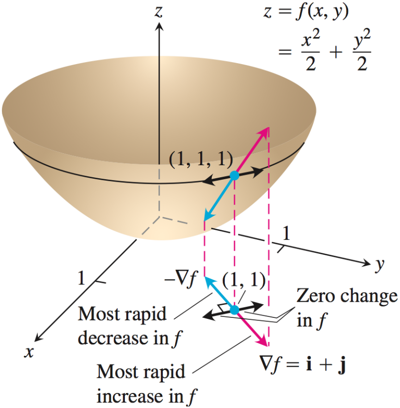
Ex: Let $f(x,y) = \dfrac{x^2}{2} + \dfrac{y^2}{2}$, $P(1,1)$ & $\mathbf{u} = \left[\frac{1}{2}, \frac{\sqrt{3}}{2}\right]$

Compute $D_{\mathbf{u}}f(P)$

$\nabla f = [x,y]$ and since $|\mathbf{u}| = 1$

$$\implies D_{\mathbf{u}}f = [x,y] \cdot \left[\frac{1}{2}, \frac{\sqrt{3}}{2}\right] = \frac{1}{2}x + \frac{\sqrt{3}}{2}y$$

$$\implies D_{\mathbf{u}}f(P) = \frac{1}{2}(1) + \frac{\sqrt{3}}{2}(1) = \frac{1+\sqrt{3}}{2}$$



Ex: Is there a direction $\mathbf{u}$ in which the rate of change of the temperature function $T(x, y) = x^2 - 3xy + 4y^2$ at $P(1, 2)$ equals $14°C/ft$? Give reasons for your answer.

ANSWER: No

REASON:

$$\nabla T = [2x - 3y, -3x + 8y] \implies \nabla T(P) = [2(1) - 3(2), -3(1) + 8(2)] = [-4, 13]$$

$$\implies |\nabla T(P)| = \|[-4, 13]\| = \sqrt{(-4)^2 + (13)^2} = \sqrt{185} \approx 13.6$$

$\implies$ Max rate of change at P is: $\sqrt{185}$

$\implies$ Min rate of change at P is: $-\sqrt{185}$

Therefore, since 14 does not lie within $\left[\sqrt{185}, -\sqrt{185}\right]$, the rate of change of the temperature can never be equal to $14°C/ft$